

Reproducible Genomics at RCAC

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Rosen Center for
Advanced Computing

Genomics Exchange Series

A discussion series, not a tool tutorial

Format: Forty minutes, then open discussion

- Intuition and strategy over step-by-step
- Bring your own broken workflow
- Questions in chat any time

Schedule: Eight sessions, Sep 15 to Dec 8

- Tuesdays, 11:00 AM ET
- Dates are uneven, check the page
- Two sessions from Tufts University

How to get value: Come back more than once

- Each session builds on the last
- Materials posted after each
- Tell me what to cover next

Four layers

Reproducibility is four separate problems, not one

Code: What you wrote

Version controlled, or not
Paths relative, or not
Committed before the run

Environment: What you loaded

- Module versions pinned
- Container image tagged
- Recorded in the job log

Data: What went in

- Reference build identified
- Inputs checksummed
- Stored where it survives

Compute: How it ran

- Thread count can change results
- Seeds set where applicable
- Resource request recorded

Demo

From fragile to reproducible, on Gautschi

Where this goes

The rest of the series

Sep 29: Data management for biologists

Oct 20: Nextflow fundamentals (Yucheng Zhang, Tufts)

Oct 27: Running nf-core pipelines on clusters (Yucheng Zhang, Tufts)

Nov 10: Nextflow, script to pipeline

Nov 17: nf-core on Open OnDemand

Dec 1: R/RStudio and r-Bioconductor

Dec 8: AI for biologists

Your turn

Two requests

Fill the survey: Under a minute, three questions:

- https://purdue.ca1.qualtrics.com/jfe/form/SV_3gax7dKKXAxtQa2

Bring a broken workflow: Any session, any time

- Post it in chat
- Or email me
- Real examples make better sessions

THANK YOU!

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Why join?

- Engage with bioinformatics experts and enthusiasts.
- Access resources, tutorials, and Q&A support.
- Participate in discussions on bioinformatics tools.



Let's grow and innovate together in the bioinformatics space!